A DERIVED COMPLETE MODULE THAT IS POWER TORSION FOR A FINITELY GENERATED IDEAL IS ANNIHILATED BY A POWER OF THAT IDEAL

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Allowed standard prerequisites (editorial).

- Derived completeness is preserved by restriction along a ring map carrying the ideal back, so it can be transferred to a polynomial algebra. invoked at proof lines 27024-27026. Lets the proof replace the ring by the polynomial ring in one variable and assume that the generator is a nonzerodivisor.
- Every module sits in a two-term sequence whose terms are adically complete modules with no torsion along the generator. invoked at proof lines 27031-27034. Provides the surjection from the complete module L whose kernel K is also complete and torsion free, which carries the whole argument.
- In a complete topological module, a countable union of closed subsets covering the whole module has a member with interior. invoked at proof lines 27045-27047. Makes one of the closed pieces N_n open in L, the key step that lets the argument climb from K to L.
- An open continuous surjection of complete topological modules has a controlled image, so a power of the generator lands inside the image. invoked at proof lines 27055-27058. Produces the exponent t with the t-th power of f contained in the image of u, which is the stated annihilation.

Source reference. Original lemma 0924: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26616–26616. Invoked at proof lines 27024-27026 to see that M is derived complete as a module over the polynomial ring with respect to the principal ideal generated by x; the use of that transfer, namely the reduction to the case of a nonzerodivisor, is inside the counted span.

Source reference. Original section 09AT: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26844–26844. Cited at proof lines 27031-27034 for the existence of the sequence with K and L adically complete and torsion free, and the footnote re-proves the only part that is used; every later step of the argument is inside the counted span.

Source reference. Original lemma 0CQV: sources/stacks-more-algebra/pinned-original-source.tex, source lines 9078–9078. Applied at proof lines 27045-27047 to the covering of L by the closures N_n to obtain an index n with N_n open; the construction of the increasing pieces L_n and the containments verified around it are inside the counted span.

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Source reference. Original lemma 0CQW: sources/stacks-more-algebra/pinned-original-source.tex, source lines 9090–9090. Used at proof lines 27055-27058 to conclude that the map u is open from the openness of the closures of the images; establishing that openness by the continuity and openness of multiplication by a power of f is inside the counted span.

Lemma 0.2 (Bhatt)**.** *Let I be a finitely generated ideal in a ring A. Let M be a derived complete A-module. If M is an I-power torsion module, then $I^nM = 0$ for some n.*

Proof. Say $I = (f_1, \ldots, f_r)$. It suffices to show that for each i there is an n_i such that $f_i^{n_i}M = 0$. Hence we may assume that $I = (f)$ is a principal ideal. Let $B = \mathbf{Z}[x] \to A$ be the ring map sending x to f. By Lemma 0924 we see that M is derived complete as a B-module with respect to the ideal (x). After replacing A by B, we may assume that f is a nonzerodivisor in A.

Assume $I = (f)$ with $f \in A$ a nonzerodivisor. According to Example 09AT there exists a short exact sequence

$$0 \to K \xrightarrow{u} L \to M \to 0$$

where K and L are I-adically complete A-modules whose f-torsion is zero[1]. Consider K and L as topological modules with the I-adic topology. Then u is continuous. Let

$$L_n = \{x \in L \mid f^n x \in u(K)\}$$

Since M is f-power torsion we see that $L = \bigcup L_n$. Let N_n be the closure of L_n in L. By Lemma 0CQV we see that N_n is open in L for some n. Fix such an n. Since $f^{n+m} : L \to L$ is a continuous open map, and since $f^{n+m}L_n \subset u(f^mK)$ we conclude that the closure of $u(f^mK)$ is open for all $m \geq 1$. Thus by Lemma 0CQW we conclude that u is open. Hence $f^tL \subset \mathrm{Im}(u)$ for some t and we conclude that f^t annihilates M as desired. □

[1]For the proof it is enough to show that there exists a sequence $K \xrightarrow{u} L \to M \to 0$ where K and L are I-adically complete A-modules. This can be shown by choosing a presentation $F_1 \to F_0 \to M \to 0$ with F_i free and then setting K and L equal to the f-adic completions of F_1 and F_0. Namely, as f is a nonzerodivisor these completions will be the derived completions and the sequence will remain exact.